

Course Folder Management System

Email (SMTP) Configuration Guide - invitations and password resets

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The system sends email in exactly two situations: (1) when the focal person **invites a faculty member** (the email contains a one-time link where the person sets their own password), and (2) when someone uses **Forgot password**. No other email is ever sent, so the volume is very small - a few messages per semester.

1. How it works without any configuration

If you deploy the system without filling in the email settings, it does not break. Instead of sending, the full email (including the invite link) is printed to the application log. To read it on the server:

```
docker compose -f docker-compose.prod.yml logs web | tail -40
```

Copy the link that starts with `http://...:8000/invite/` and give it to the faculty member (for example over WhatsApp or in person). This works fine, but real email is more convenient - pick one of the three options below.

2. The settings the application reads

All email configuration goes into the `.env` file on the server (the same file created during deployment). These are the lines that matter:

```
EMAIL_HOST= # SMTP server name, e.g. smtp.gmail.com EMAIL_PORT=587 # leave 587 unless told otherwise
EMAIL_HOST_USER= # the SMTP login (usually an email address) EMAIL_HOST_PASSWORD= # the SMTP password / app
password / key EMAIL_USE_TLS=True # leave True DEFAULT_FROM_EMAIL=Course Folder System <no-reply@uiit.edu.pk>
```

After editing `.env`, apply the change by recreating the web container:

```
docker compose -f docker-compose.prod.yml up -d --force-recreate web
```

3. Option A - university mail server (preferred if available)

If UIIT / PMAS-AAUR has its own mail server, this is the cleanest option because emails come from a real `uiit.edu.pk` address and are trusted by recipients.

1. Ask the university IT / network section for: the SMTP server name, the port (usually 587), and a mailbox the system may send from (for example `no-reply@uiit.edu.pk` or the focal person's mailbox) with its password.

Fill the values into `.env`:

```
EMAIL_HOST=mail.uaar.edu.pk # whatever IT gives you EMAIL_PORT=587 EMAIL_HOST_USER=no-reply@uiit.edu.pk
EMAIL_HOST_PASSWORD=the-mailbox-password EMAIL_USE_TLS=True DEFAULT_FROM_EMAIL=Course Folder System
<no-reply@uiit.edu.pk>
```

3. Recreate the container (command in section 2) and send a test invite.

4. Option B - a Gmail account with an App Password (quickest)

Any Gmail account can send the system's mail. Google no longer allows your normal account password over SMTP - you must create a 16-character **App Password**, and that requires **2-Step Verification** to be turned on first.

4.1 Create the App Password (in a browser, once)

1. Sign in to the Gmail account you want to send from (a dedicated account such as `uiit.coursefolders@gmail.com` is tidier than a personal one).
2. Go to `myaccount.google.com/security` and turn on **2-Step Verification** (phone number confirmation).

- Go to `myaccount.google.com/apppasswords`, enter a name such as "Course Folder System", and press Create.
- Google shows a 16-character password once. Copy it (spaces do not matter - remove them).

4.2 Put it in .env

```
EMAIL_HOST=smtp.gmail.com EMAIL_PORT=587 EMAIL_HOST_USER=uiit.coursefolders@gmail.com
EMAIL_HOST_PASSWORD=abcdefghijklmnop # the 16-character app password EMAIL_USE_TLS=True DEFAULT_FROM_EMAIL=Course
Folder System <uiit.coursefolders@gmail.com>
```

Note: with Gmail, `DEFAULT_FROM_EMAIL` must use the same Gmail address you log in with - Gmail rewrites the sender otherwise. Free Gmail allows roughly 500 recipients per day, far more than this system ever needs.

5. Option C - Brevo free tier (no personal account involved)

Brevo (formerly Sendinblue) has a free plan with **300 emails per day**, which is more than enough. It does not require a domain of your own for basic use, but each sender address must be verified.

- Create a free account at `brevo.com` and confirm your email.
- In the dashboard, add and verify a **sender**: Senders & IP - Senders - Add a sender. Brevo emails a 6-digit code to that address; enter it. Use an address you can actually receive mail on (the focal person's email works).
- Go to the **SMTP & API** page and press **Generate a new SMTP key**. Copy the *login* shown there and the *key* (the key is shown only once - save it). Make sure it is an SMTP key, not an API key.

Fill `.env`:

```
EMAIL_HOST=smtp-relay.brevo.com EMAIL_PORT=587 EMAIL_HOST_USER=the-smtp-login-shown-by-brevo
EMAIL_HOST_PASSWORD=the-smtp-key EMAIL_USE_TLS=True DEFAULT_FROM_EMAIL=Course Folder System
<the-verified-sender-address>
```

Important: `DEFAULT_FROM_EMAIL` must be one of the senders you verified in step 2, or Brevo silently drops / rejects the message.

6. Test it end to end

- Apply the settings: `docker compose -f docker-compose.prod.yml up -d --force-recreate web`
- Quick check from the server (sends a real test message):

```
docker compose -f docker-compose.prod.yml exec web \ python manage.py shell -c "from django.core.mail import
send_mail; \ send_mail('Test - Course Folder System', 'Email works.', None, ['YOUR-OWN-EMAIL'])"
```

If this prints `1` and the message arrives (check spam too), email works.

- Real check: log in as the focal person, invite a faculty member with a real address, and confirm the invitation arrives and the link opens the set-password page.

7. Troubleshooting

Symptom	Likely cause and fix
"The invitation email could not be sent" message in the app	Wrong host, port, username, or password in <code>.env</code> . The account was NOT created, so after fixing the settings simply invite again. Details are in <code>docker compose -f docker-compose.prod.yml logs web</code> .
Authentication error with Gmail	You used the normal account password. Only a 16-character App Password works, and 2-Step Verification must be on.
Connection timeout	The university network/firewall may block outgoing port 587. Ask IT to allow it from the server, or use the university mail server (Option A).
Message never arrives	Check the recipient's spam folder first. With Brevo, confirm the sender address is verified and matches <code>DEFAULT_FROM_EMAIL</code> .

No email settings at all	Not an error - invite links are printed to the log (section 1).
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8. Which option should you pick?

Option	Best when	Cost	Effort
A - University SMTP	IT can give you a mailbox; most professional sender address	Free	One email to IT
B - Gmail App Password	You want it working today with zero approvals	Free	10 minutes
C - Brevo free tier	You prefer a neutral service and delivery statistics	Free (300/day)	15 minutes

Recommendation: start with **Option B (Gmail)** so invitations work immediately, and switch to **Option A** when the university mailbox is arranged - it is only a four-line change in `.env`.